1) Scope

The following document provides instructions on how to modify the MakerWheel master file and use it to generate custom MakerWheel designs. The MakerWheel master file contains an M8 MakerWheel with 64 teeth (64T) as a starting point (henceforth referred to as the "Baseline MakerWheel"). Custom MakerWheels can be generated by modifying certain parameters of the Baseline MakerWheel. The master file is highly dependent on equation driven dimensioning, which reduces the number of user inputs required. Instructions are provided below to make the following modifications:

- Procedure A:Change the number of MakerWheel teeth.
- Procedure B: Change the diameter and pitch of the mating screw thread.

2) Legal

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3) Prerequisites

a) Access to a computer with SolidWorks® 2016 (or newer) software.

b) A basic knowledge of how to use SolidWorks® software.

4) Limitations

a) The MakerWheel master file screw thread profile is compatible with screw threads that have a 60 degree thread angle and V-shaped profile. This includes ISO metric and Unified Thread Standard (UTS) screws. Unified Coarse (UNC) and Unified Fine (UNF) are part of the UTS standard and are compatible. For other thread types, such as Acme or metric trapezoidal, the geometry of the thread profile must be modified. This is possible but is outside the scope of this document.

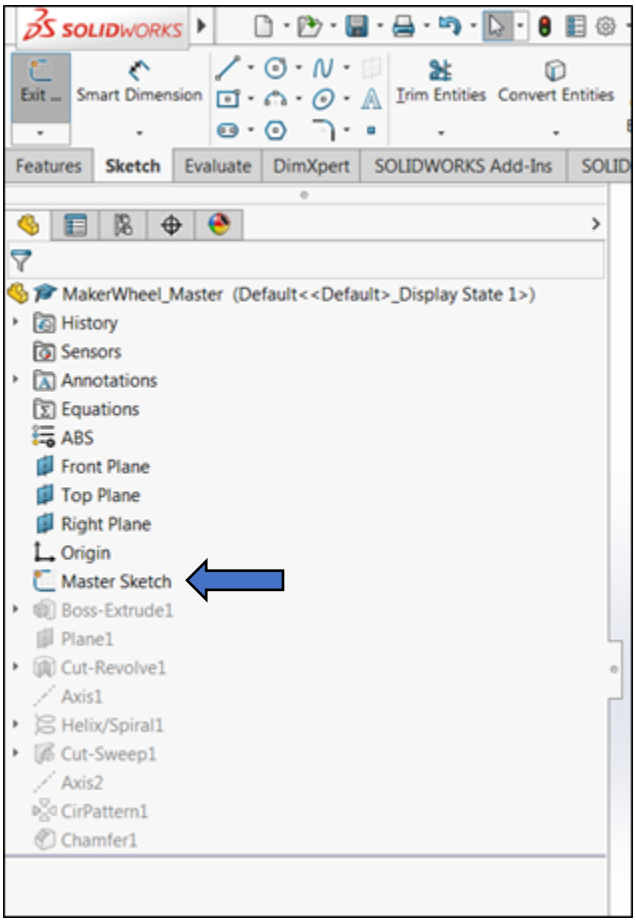
5) Procedure A: Change the number of MakerWheel teeth

Note: Follow this procedure if you want to create an M8 MakerWheel with a different number of teeth. If you want to create a MakerWheel that has a different number of teeth and mates with a different screw size or pitch, follow Procedure B in section 6.

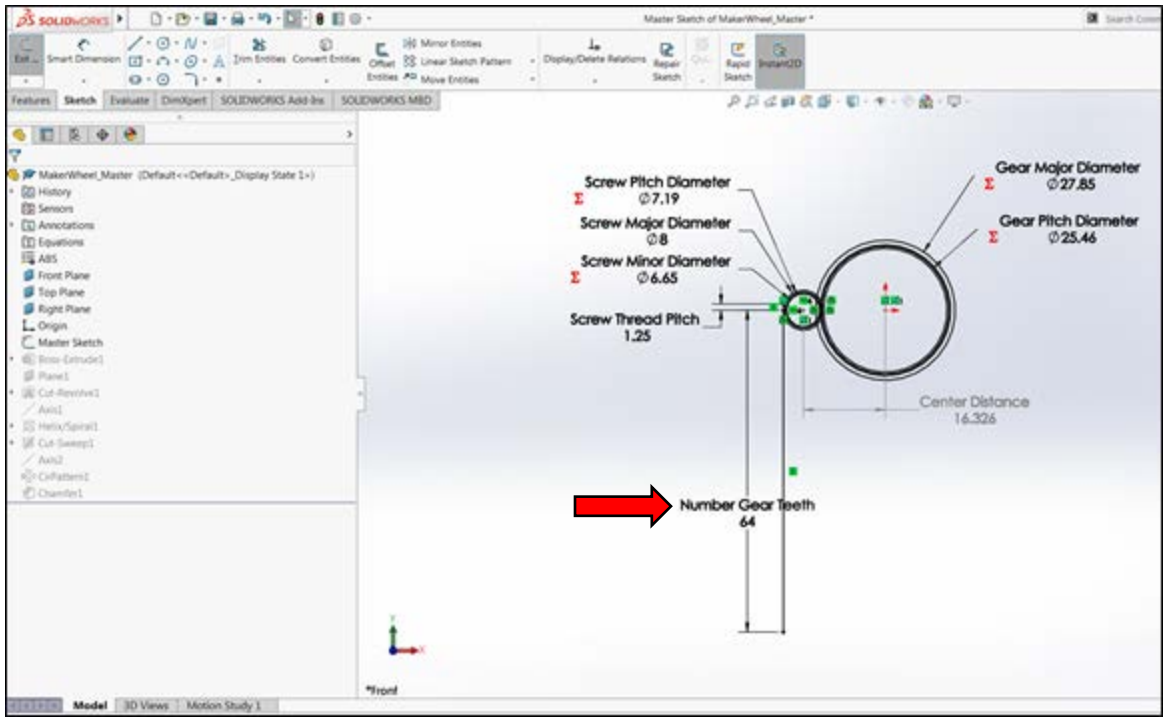
a) Open the MakerWheel master file in SolidWorks®.

WARNING: SOLIDWORKS® UNITS MUST BE MILIMETERS FOR DIMENSIONS TO BE CORRECT. VERIFY UNIT SYSTEM IS MMGS (MILIMETERS GRAMS SECONDS).

b) Open the sketch called "Master Sketch" (identified by the blue arrow in the figure below) by right clicking on it and selecting the edit sketch icon .



c) Locate the dimension called “Number Gear Teeth” (identified by the red arrow in the figure below).



d) Double click on the dimension value to edit it and enter a new numerical value between 8 and 100 (whole numbers only, numbers outside this range have not been verified and may not regenerate properly).

Note 1: Changing this value should cause the equation driven dimensions “Gear Major Diameter” and “Gear Pitch Diameter” to change as well.

Note 2: Additional modifications are necessary for teeth values greater than 100. The number of revolutions of the “Helix/Spiral1” feature must be increased. See Procedure B, Section 6i) through j) below for instructions on how to do this.

e) Accept the changes and exit the sketch. Manually rebuild the model if necessary by clicking on the rebuild icon . Note that long rebuild times may be experienced depending on your computer hardware.

f) The procedure is complete and you should now have an M8 MakerWheel with a modified number of teeth.

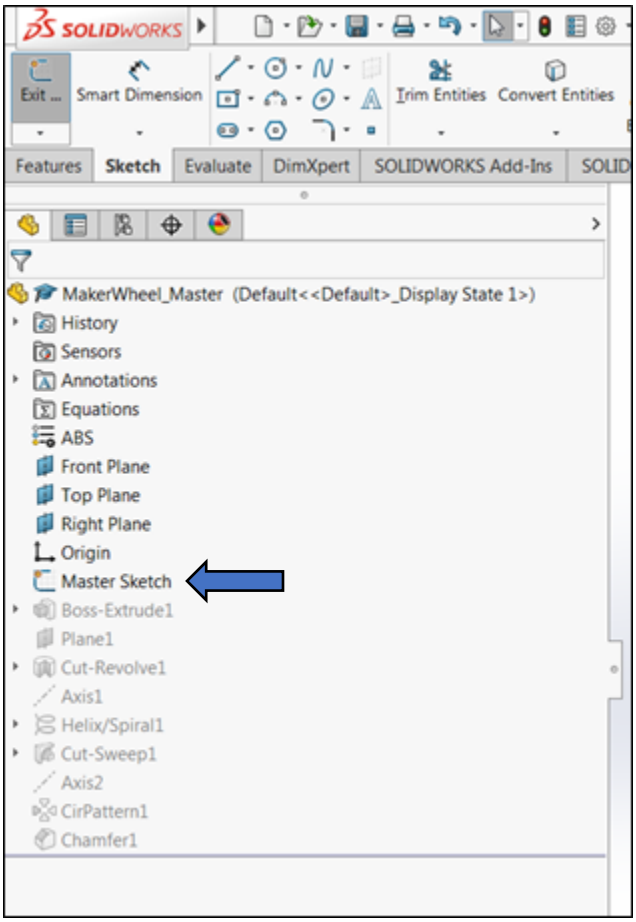
6) Procedure B: Change the diameter and pitch of the mating screw thread.

Note: Follow this procedure if you want to create a MakerWheel that mates with a different screw size or pitch (changing the number of teeth is optional). If you just want an M8 MakerWheel with a different number of teeth, follow Procedure A in section 5.

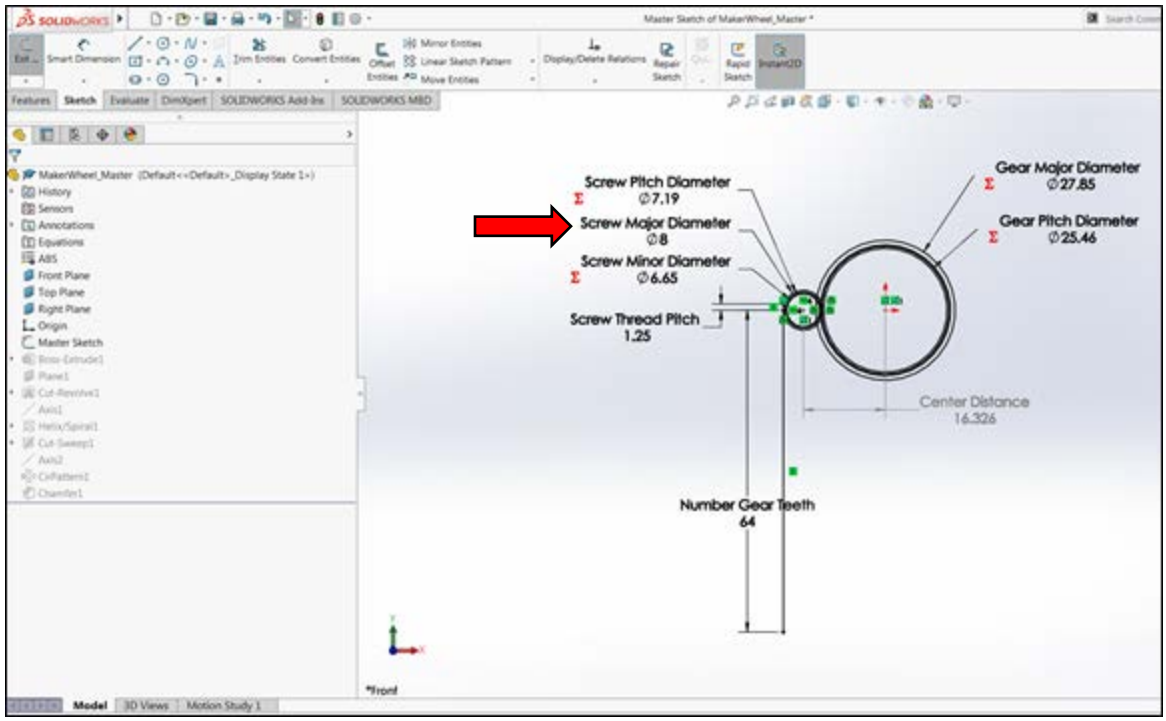
a) Open the MakerWheel master file in SolidWorks®.

WARNING: SOLIDWORKS® UNITS MUST BE MILIMETERS FOR DIMENSIONS TO BE CORRECT. VERIFY UNIT SYSTEM IS MMGS (MILIMETERS GRAMS SECONDS) BEFORE PROCEEDING.

b) Open the sketch called "Master Sketch" (identified by the blue arrow in the figure below) by right clicking on it and selecting the edit sketch icon.



c) Locate the dimension called “Screw Major Diameter” (identified by a red arrow in the figure below).



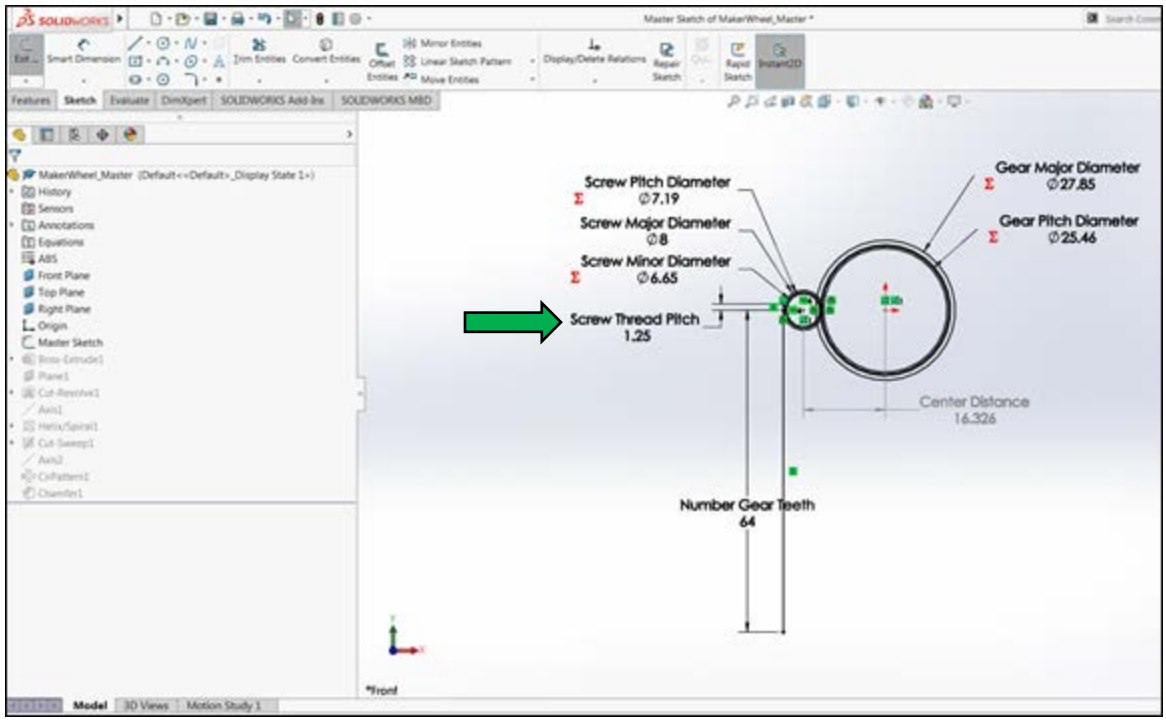
d) Double click on the dimension value to edit it and enter the numerical value of the new screw diameter in millimeters.

i) Example 1: You have an ISO metric M10x1.5 threaded rod or screw. A value of “10” would be entered since the major diameter is 10 mm.

ii) Example 2: You have a UNC ½”-13 threaded rod or screw. A value of “12.7” would be entered since the major diameter is 12.7 mm (0.5 inch converted to mm).

Note: Changing this value should cause the equation driven dimensions “Screw Pitch Diameter” and “Screw Minor Diameter” to change as well.

e) Locate the dimension called “Screw Thread Pitch” (identified by a green arrow in the figure below).



f) Double click on the dimension value to edit it and enter the numerical value of the new screw thread pitch in millimeters.

i) Example 1: You have an ISO metric M10x1.5 threaded rod or screw. A value of “1.5” would be entered since the thread pitch is 1.5 mm.

ii) Example 2: You have a UNC ½”-13 threaded rod or screw. UNC and UNF threads are given in “threads per inch” rather than pitch and must be converted. The number “13” in ½”-13 indicates that the thread has 13 threads per inch. The pitch is calculated as:

[(1 inch) / (13 threads per inch)] x 25.4 mm/inch = 1.95 mm

Note: To avoid rounding errors, it is best to input the conversion equation as the dimension value and let SolidWorks® calculate the pitch value for you.

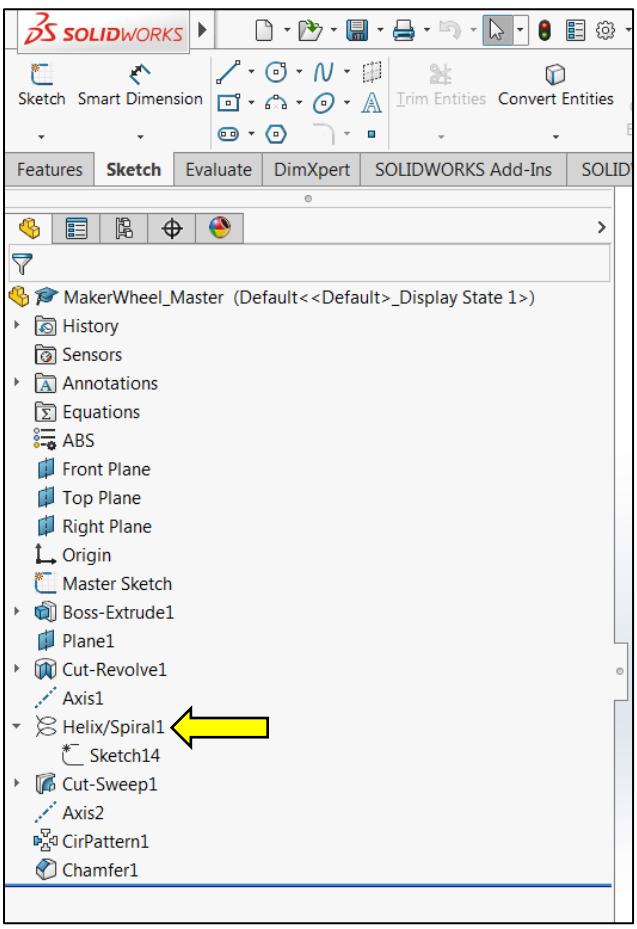
g) If desired, the number of MakerWheel teeth should be changed at this time. Double click on the dimension called “Number Gear Teeth” to edit it and enter a new numerical value between 8

and 100 (whole numbers only, numbers outside this range have not been verified and may not regenerate properly).

h) Accept all changes and exit the sketch. Manually rebuild the model if necessary by clicking on the rebuild icon . Note that long rebuild times may be experienced depending on your computer hardware.

Note: The pitch value must be changed manually in one other location. The rebuild may fail because this other pitch value has not been changed yet. This is okay.

i) Open the feature called "Helix/Spiral1" (identified by a yellow arrow in the figure below) by right clicking on it and selecting the edit feature icon .

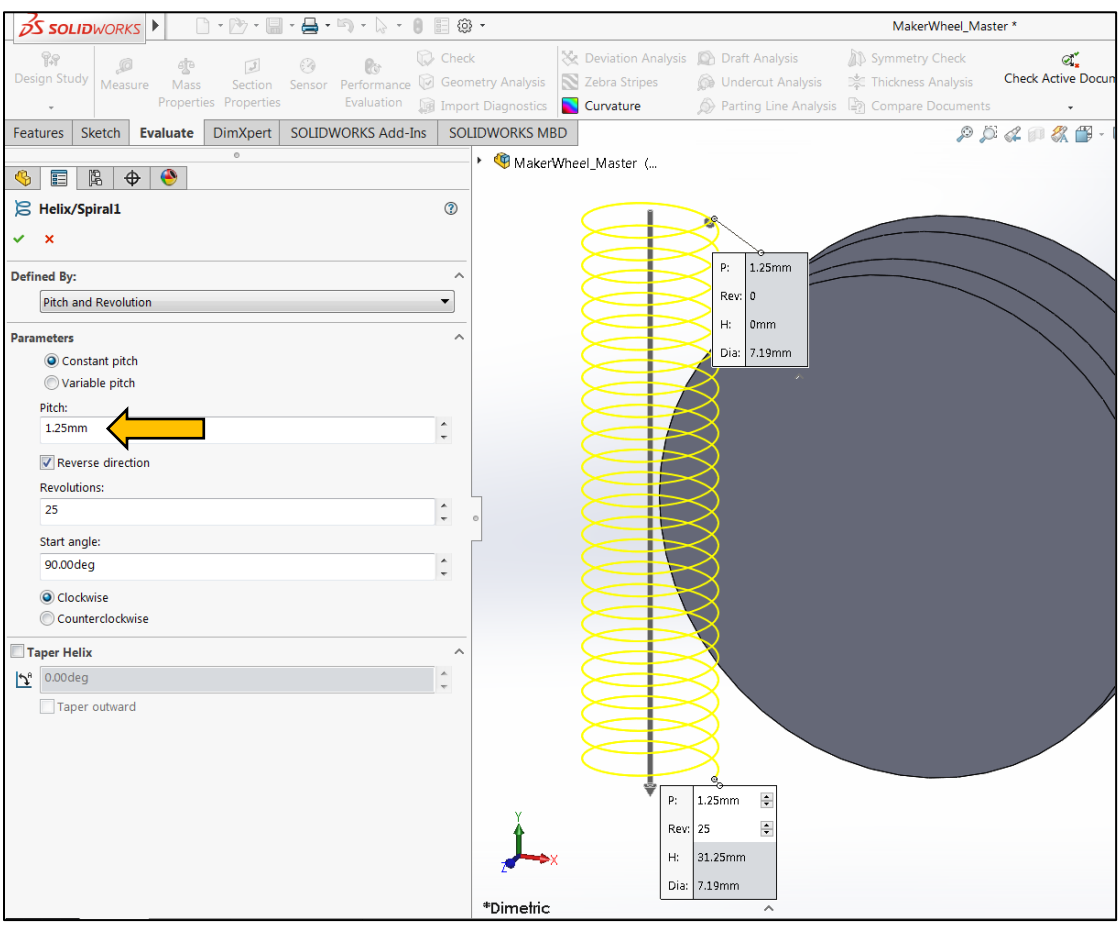


j) Locate the pitch input field (identified by an orange arrow in the figure below) and enter the same pitch value that you used previously in step 6f).

Note 1: The number of helix revolutions is set to 25 by default. This value may be decreased to a smaller value for smaller diameter MakerWheels to improve rebuild times but must extend at least one revolution past the last point where the helix makes contact with the body of the MakerWheel.

Note 2: The number of helix revolutions will need to be increased for large MakerWheels. Ensure that the helix extends at least one revolution past the last point where the helix makes contact with the body of the MakerWheel.

Note 3: SolidWorks® 2016 does not allow the pitch value in the helix feature to be linked to the pitch value in the Master Sketch using an equation. Later versions of SolidWorks® may allow this. Please report this feature if it is available on later versions of SolidWorks®.



k) Accept all changes and exit the feature. Manually rebuild the model if necessary by clicking on the rebuild icon . Note that long rebuild times may be experienced depending on your computer hardware.

l) The procedure is complete and you should now have a MakerWheel that will mate with a different thread diameter and/or pitch.